Output

2) ii)

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STUDENT01	USN	Varchar2	10	-	-	1	-	-	-
	NAME	Varchar2	50	-	-	-	-	-	-
	DOB	Date	7	-	-	-	✓	-	-
	BRANCH	Varchar2	30	-	-	-	✓	-	-
	M1	Number	-	3	0	-	✓	-	-
	M2	Number	-	3	0	-	✓	-	-
	M3	Number	-	3	0	-	✓	-	-
	TOTAL	Number	-	10	0	-	✓	-	-
	GPA	Number	-	8	2	-	✓	-	-

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3) iv)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1001	Sujal	10-JAN-04	CSA	90	80	95	-	-
1002	Musthaq	21-OCT-04	BCA	89	90	92	-	-
1003	Gagan	06-DEC-03	MCA	90	93	97	-	-
1004	Eshwar	01-FEB-03	MBA	93	86	82	-	-
1005	Prajwal	17-APR-04	BBA	91	79	85	-	-
1006	Sneha	18-JUL-04	BBA	90	89	85	-	-

Date:

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Consider the following schema:
STUDENT (USN, name, date_of_birth, branch, mark1, mark2, mark3, total, GPA)

Perform the following

1. Creating Tables (with and Without constraints), Inserting/Updating/Deleting Records in a Table, Saving (Commit) and Undoing (rollback)

i)
```
create table student01(
USN varchar2(10) constraint ss1 primary key,
Name varchar2(50) constraint ss2 not null,
DOB date, Branch varchar2(30),
m1 number(3) constraint ss3 check (m1>=0),
m2 number(3) constraint ss4 check (m2>=0),
m3 number(3) constraint ss5 check (m3>=0),
Total number(10),
GPA number(8,2));
```

ii)
```
describe student01;
```

iii)
```
INSERT INTO STUDENT01(USN, Name, DOB, Branch, m1, m2, m3)
VALUES('1001','Sujal','10-Jan-2004','CSA', 90, 80, 95);
INSERT INTO STUDENT01(USN, Name, DOB, Branch, m1, m2, m3)
VALUES('1002','Musthaq','21-Oct-2004','BCA', 89, 90, 92);
INSERT INTO STUDENT01(USN, Name, DOB, Branch, m1, m2, m3)
VALUES('1003','Gagan','6-Dec-2003','MCA', 90, 93, 97);
INSERT INTO STUDENT01(USN, Name, DOB, Branch, m1, m2, m3)
VALUES('1004','Eshwar','1-Feb-2003','MBA', 93, 86, 82);
```

Teacher's Signature

1) vi)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1001	Sujal	10-JAN-04	CSA	90	80	95	—	—
1002	Musthaq	21-OCT-04	BCA	89	90	92	—	—
1003	Gagan	06-DEC-03	MCA	90	93	97	—	—
1004	Eshwar	01-FEB-03	MBA	93	86	82	—	—
1005	Prajwal	17-APR-04	BBA	91	79	85	—	—
1006	Sneha	18-JUL-04	BBA	95	89	85	—	—

1) viii)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1001	Sujal	10-JAN-04	CSA	90	80	95	—	—
1002	Musthaq	21-OCT-04	BCA	89	90	92	—	—
1003	Gagan	06-DEC-03	MCA	90	93	97	—	—
1004	Eshwar	01-FEB-03	MBA	93	86	82	—	—
1005	Prajwal	17-APR-04	BBA	91	79	85	—	—

4a)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1001	Sujal	10-JAN-04	CSA	90	80	95	—	—

4b)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1001	Sujal	10-JAN-04	CSA	90	80	95	265	—
1002	Musthaq	21-OCT-04	BCA	89	90	92	271	—
1003	Gagan	06-DEC-03	MCA	90	93	97	280	—
1004	Eshwar	01-FEB-03	MBA	93	86	82	261	—
1005	Prajwal	17-APR-04	BBA	91	79	85	255	—

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```
INSERT INTO STUDENT01(USN, Name, DOB, Branch, m1, m2, m3)
VALUES ('1005', 'Prajwal', '17-Apr-2004', 'BBA', 91, 79, 85);
INSERT INTO STUDENT01(USN, Name, DOB, Branch, m1, m2, m3)
VALUES ('1006', 'Sneha', '18-Jul-2004', 'BBA', 90, 89, 85);
Select * from Student01;
```

iv) `UPDATE Student01 set m1=95 WHERE USN='1006';`

vi) `Select * from Student01;`

vii) `DELETE FROM Student01 WHERE USN='1006';`

viii) `Select * from Student01;`

4) Execute the following queries:

a. Find the students whose name starts with the alphabet "S".

```
Select * from student01 where name like 'S%';
```

b. Update the column total by adding the columns mark1, mark2, mark3.

```
UPDATE Student01 SET TOTAL = m1+m2+m3;
Select * from Student01;
```

Teacher's Signature

2a)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1001	Sujal	10-JAN-04	CSA	90	80	95	265	88.33
1002	Musthaq	21-OCT-04	BCA	89	90	92	271	90.33
1003	Gagan	06-DEC-03	MCA	90	93	97	280	93.33
1004	Eshwar	01-FEB-03	MBA	93	86	82	261	87
1005	Prajwal	17-APR-04	BBA	91	79	85	255	85

2b)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1003	Gagan	06-DEC-03	MCA	90	93	97	280	93.33
1004	Eshwar	01-FEB-03	MBA	93	86	82	261	87

3a)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1001	Sujal	10-JAN-04	CSA	90	80	95	265	88.33

3b)

BRANCH	MAX(GPA)
MBA	87
BCA	90.33
MCA	93.33
CSA	88.33
BBA	85

5a)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1004	Eshwar	01-FEB-03	MBA	93	86	82	261	87

Date:

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2a. Find the GPA score of all the students.

```
UPDATE student01 set GPA = total/3;

Select * from student01;
```

b. Find the students who born on a particular year of birth from the date of birth column.

```
SELECT * FROM student01 WHERE TO_CHAR(DOB,'YYYY')
LIKE '%2003%';
```

3a. List the students who are studying in a particular branch of study.

```
Select * from student01 where Branch = 'CSA';
```

b. Find the maximum GPA score of the student branch-wise.

```
Select branch, max(GPA) from student01 group by branch;
```

5a. Find the students whose name ends with the alphabets "AR".

```
Select * from student01 where name like '%ar';
```

Teacher's Signature

5b)

USN	NAME	DOB	BRANCH	M1	M2	M3	TOTAL	GPA
1002	Musthaq	21-OCT-04	BCA	89	90	92	271	90.33
1003	Gragan	06-DEC-03	MCA	90	93	97	280	93.33
1004	Eshwar	01-FEB-03	MBA	93	86	82	261	87
1005	Projwal	17-APR-04	BBA	91	79	85	255	85

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5b. Delete the student details whose USN is given as 1001

```
delete from student01 where USN = '1001';

select * from student01;
```

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Output

1)a)iii)

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
BRANCH001	BRANCH_ID	Number	–	4	0	1	–	–	–
	BRANCH_NAME	Varchar2	30	–	–	–	–	–	–
	HOD	Varchar2	40	–	–	–	✓	–	–

1-3

1)a)iv)

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
AUTHOR001	AUTHOR_ID	Number	–	4	0	1	–	–	–
	AUTHOR_NAME	Varchar2	30	–	–	–	–	–	–
	COUNTRY	Varchar2	20	–	–	–	✓	–	–
	AGE	Number	–	3	0	–	✓	–	–

1-4

1)a)vi)

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STUDENT001	USN	Varchar2	10	–	–	1	–	–	–
	NAME	Varchar2	20	–	–	–	–	–	–
	ADDRESS	Varchar2	30	–	–	–	✓	–	–
	BRANCH_ID	Number	–	4	0	–	✓	–	–
	SEM	Number	–	2	0	–	✓	–	–

1-5

Date:

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Create the following tables with properly specifying Primary keys, Foreign keys and solve the following queries

BRANCH (Branchid, Branchname, HOD)
STUDENT (USN, Name, Address, Branchid, sem)
BOOK (Bookid, Bookname, Authorid, Publisher, Branchid)
AUTHOR (Authorid, Authorname, Country, Age)
BORROW (USN, Bookid, Borrowed_Date).

1. Perform the following:

a. Viewing all databases, Creating a Database, Viewing all Tables in a Database, Creating Tables (with and without constraints), Inserting/Updating/Deleting Records in a Table, Saving (commit) and Undoing (rollback).

i
```
create table branch001(
Branch_Id number(4) constraint b1 primary key,
Branch_Name varchar(30) constraint b2 not null,
HOD varchar2(40));
```

ii
```
desc branch001;
```

iii
```
create table author001(
Author_Id number(4) constraint a1 primary key,
Author_Name varchar2(30) constraint a2 not null,
Country varchar2(20),
Age number(3));
```

iv
```
desc author001;
```

v
```
Create table student001(
USN varchar2(10) constraint s1 primary key,
Name varchar2(20) constraint s2 not null,
```

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1) a) viii)

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
Book001	BOOKID	Number	-	4	0	1	-	-	-
	BOOK_NAME	Varchar2	40	-	-	-	-	-	-
	AUTHOR_ID	Number	-	4	0	-	✓	-	-
	PUBLISHER	Varchar2	30	-	-	-	✓	-	-
	BRANCH_ID	Number	-	4	0	-	✓	-	-

1-5

1) a) x)

Table	Column	Data Type	Length	Precision	Scale	Primary key	Nullable	Default	Comment
BORROW001	USN	Varchar2	10	-	-	1	-	-	-
	BOOK_ID	Number	-	4	0	2	-	-	-
	BORROWED_DATE	Date	7	-	-	-	✓	-	-

1-3

xi)

BRANCH_ID	BRANCH_NAME	HOD
1	MCA	Vineeta
2	MBA	Nagarjuna
3	Science	Nagendra
4	Commerce	Vaishnavi
5	CSA	Sneha

Date:

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```
    Address varchar2(30),
    Branch_Id number(4) constraint s3 references branch001(Branch_Id),
    sem number(2));
vi  desc student001;

vii create table book001(
    Book_Id number(4) constraint bk1 primary key,
    Book_Name varchar2(40) constraint bk2 not null,
    Author_Id number(4) constraint bk3 references author001(Author_Id),
    Publisher varchar2(30),
    Branch_Id number(4) constraint bk4 references branch001(Branch_Id));
viii desc book001;

ix  create table borrow001(
    USN varchar2(10) constraint bk61 references student002(USN),
    Book_Id number(4) constraint bk62 references book002(Book_Id),
    Borrowed_Date date,
    constraint bk63 primary key(USN, Book_Id));
x   desc borrow001;

    INSERT INTO BRANCH001 VALUES(1,'MCA','Vineeta');
    INSERT INTO BRANCH001 VALUES(2,'MBA','Nagarjuna');
    INSERT INTO BRANCH001 VALUES(3,'Science','Nagendra');
    INSERT INTO BRANCH001 VALUES(4,'Commerce','Vaishnavi');
    INSERT INTO BRANCH001 VALUES(5,'CSA','Sneha');
xi  select * from branch001;

    INSERT INTO AUTHOR001 VALUES(101,'Salman Rushdie','India',40);
    INSERT INTO AUTHOR001 VALUES(102,'Jane Austen','England',30);
```

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xii)

AUTHOR_ID	AUTHOR_NAME	COUNTRY	AGE
101	Salman Rushdie	India	40
102	Jane Austen	England	30
103	Paulo Coelho	Brazil	55
104	Haruki Murakami	Japan	42
105	Fyodor Dostoevsky	Russia	38

xiii)

USN	NAME	ADDRESS	BRANCH_ID	SEM
SS01	Kartik	Karwar	2	1
SS02	Malthi	Honnavar	4	2
SS03	Faiha	Dandeli	5	1
SS04	Unnati	Haliyal	3	2
SS05	Sujal	West Bengal	1	2

xiv)

BOOKID	BOOK_NAME	AUTHOR_ID	PUBLISHER	BRANCH_ID
1001	Midnights Children	101	Jonathan Cape	2
1002	Pride and Prejudice	102	Penguin Classics	4
1003	The Alchemist	103	Harper Collins	3
1004	Norwegian Wood	104	Kodansha	5
1005	Crime and Punishment	105	The Russian Messenger	1

```
INSERT INTO AUTHOR001 VALUES(103,'Paulo Coelho','Brazil', 55);
INSERT INTO AUTHOR001 VALUES(104,'Haruki Murakami','Japan', 42);
INSERT INTO AUTHOR001 VALUES(105,'Fyodor Dostoevsky','Russia', 38);
```

xii
```
Select * from author001;
```

```
INSERT INTO STUDENT001 VALUES('SS01','Kartik','Kanwar', 2,1);
INSERT INTO STUDENT001 VALUES('SS02','Mathi','Honnavar', 4,2);
INSERT INTO STUDENT001 VALUES('SS03','Faiha','Dandeli', 5,1);
INSERT INTO STUDENT001 VALUES('SS04','Unnati','Haliyal', 3,2);
INSERT INTO STUDENT001 VALUES('SS05','Sujal','West Bengal', 1,2);
```

xiii
```
Select * from student001;
```

```
INSERT INTO BOOK001 VALUES(1001,'Midnights Children',101,'Jonathan
                           cape',2);
INSERT INTO BOOK001 VALUES(1002,'Pride and Prejudice',102,
                           'Penguin Classics',4);
INSERT INTO BOOK001 VALUES(1003,'The Alchemist',103,'HarperCollins',
                           3);
INSERT INTO BOOK001 VALUES(1004,'Norwegian Wood',104,
                           'Kodansha',5);
INSERT INTO BOOK001 VALUES(1005,'Crime and Punishment',105,
                           'The Russian Messenger',1);
```

xiv
```
Select * from book001;
```

```
INSERT INTO BORROW001 VALUES('SS05',1003,'10-MAY-2026');
INSERT INTO BORROW001 VALUES('SS05',1005,'11-JUN-2026');
INSERT INTO BORROW001 VALUES('SS05',1004,'16-AUG-2026');
INSERT INTO BORROW001 VALUES('SS04',1003,'15-MAY-2026');
INSERT INTO BORROW001 VALUES('SS03',1002,'10-MAY-2026');
```

xv)

USN	BOOK_ID	BORROWED_DATE
SS05	1003	10-MAY-26
SS05	1005	11-JUN-26
SS05	1004	16-AUG-26
SS04	1003	15-MAY-26
SS03	1002	10-MAY-26
SS02	1001	21-FEB-26

2a)

USN	NAME	ADDRESS	BRANCH_ID	SEM
SS05	Sujal	West Bengal	1	2

2b)

USN	NAME	ADDRESS	BRANCH_ID	SEM
SS01	Kartik	Karwar	2	1

3a)

USN	NAME	BRANCH_NAME	BOOK_NAME	AUTHOR_NAME	BORROWED DATE
SS05	Sujal	MCA	The Alchemist	Paulo Coelho	10-MAY-26
SS05	Sujal	MCA	Crime and Punishment	Fyodor Dostoevsky	11-JUN-26
SS05	Sujal	MCA	Norwegian Wood	Haruki Murakami	16-AUG-26

Date:

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INSERT INTO BORROW001 VALUES ('SS02', 1001, '21-FEB-2026');

xv Select * from borrow001;

2 a List the details of Students who are all studying in 2nd sem MCA

Select * from student001 where
sem = 1 and Branch_Id =
(Select Branch_Id from branch001 where
Branch_Name = 'MCA');

b list the students who are not borrowed any books

Select * from student001 where USN not in
(select USN from Borrow001);

3 a Display the USN, Studentname, Branch_name, Book_name, Author_name Books_Borrowed_Date of 2nd sem MCA students who borrowed books

Select student001.USN, student001.Name, branch001.Branch_Name, book001.Book_Name, author001.Author_Name, borrow001.Borrowed Date from student001, branch001, book001, author001, borrow001 where student001.Branch_Id = branch001.Branch_Id and book001.Author_Id = author001.Author_Id and borrow001.USN = student001.USN and borrow001.Book_Id = book001.Book_Id and sem = 2 and branch001.Branch_Name = 'MCA';

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3b)

AUTHOR_ID	COUNT(*)
101	1
102	1
103	1
104	1
105	1

4a)

USN	NAME	ADDRESS	BRANCH_ID	SEM
SS05	Sujal	West Bengal	1	2

4b)

USN	NAME	ADDRESS	BRANCH_ID	SEM
SS05	Sujal	West Bengal	1	2

5a)

BOOK_NAME
The Alchemist
Pride and Prejudice
Norwegian Wood
Midnights children
Crime and Punishment

5b)

USN	NAME	ADDRESS	BRANCH_ID	SEM
SS03	Sujal	West Bengal	1	2
SS04	Unnati	Haliyal	3	2

3b Display the number of books written by each Author.

```
Select Author_Id, count(*) from book001 group by Author_Id;
```

4a Display the student details who borrowed more than two books.

```
Select * from student001 where USN in
(Select USN from borrow001 group by USN having count(*)>=2);
```

4b Display the student details who borrowed books of more than one Author.

```
Select * from student001 where exists
(Select USN from borrow001, Book001 where
book001.Book_Id = Borrow001.Book_Id and
student001.USN = borrow001.USN group by
USN having count (distinct Author_Id) > 1);
```

5a Display the Book names in descending order of their names

```
Select Book_Name from Book001 order by Book_Name desc;
```

5b List the details of students who borrowed the books which are all published by the same publisher.

```
Select * from student001 where USN in
(Select USN from borrow001, book001 where
book001.Book_Id = borrow.001.Book_Id and
Publisher = 'HarperCollins');
```

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Output:

2a

Table	Column	DataType	Length	Precision	Scale	Primary key	Nullable	Default	Comment
DEPARTMENT01	DNO	Number	-	4	0	1	-	-	-
	DNAME	Varchar2	30	-	-	-	-	-	-
	MGRSSN	Number	-	4	0	-	✓	-	-
	MGRSTARTDATE	Date	7	-	-	-	✓	-	-

1-4

2b

Table	Column	DataType	Length	Precision	Scale	Primary key	Nullable	Default	Comment
EMPLOYEE01	SSN	Number	-	4	0	1	-	-	-
	FNAME	Varchar2	30	-	-	-	✓	-	-
	MINT	Varchar2	30	-	-	-	✓	-	-
	LNAME	Varchar2	30	-	-	-	✓	-	-
	BDATE	Date	7	-	-	-	✓	-	-
	ADDRESS	Varchar2	50	-	-	-	✓	-	-
	SEX	Varchar2	10	-	-	-	✓	-	-
	SALARY	Number	-	10	2	-	✓	-	-
	SUPERSSN	Number	-	4	0	-	✓	-	-
	DNO	Number	-	4	0	-	✓	-	-

1-10

Date:

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Consider the Company database with following schema
EMPLOYEE (FNAME, MINIT, LNAME, SSN, BDATE, ADDRESS, SEX, SALARY, SUPERSSN, DNO)
DEPARTMENT (DNAME, DNUMBER, MGRSSN, MGRSTARTDATE)
DEPT_LOCATIONS (DNUMBER, DLOCATION)
PROJECT (PNAME, PNUMBER, PLOCATION, DNUM)
WORKS_ON (ESSN, PNO, HOURS)
DEPENDENT (ESSN, DEPENDENT_NAME, SEX, BDATE, RELATIONSHIP)

Perform the following

2 Viewing all databases, Creating a Database, Viewing all Tables in a Database, Creating Tables (with and without Constraints), Inserting/Updating/Deleting Records in a Table, Saving (commit) and Undoing (rollback).

a. create table department01(
DNO number(4) constraint d1 primary key,
DNAME varchar2(30) constraint d2 not null,
MGRSSN number(4),
MGRSTARTDATE date);

a1 desc department01;

b Create table employee01(
SSN number(4) constraint e1 primary key,
FNAME varchar2(30),
MINT varchar2(30),
LNAME varchar2(30),
BDATE date,
ADDRESS varchar2(50),

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2c.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPTLOCATIONS01	DNO	Number	-	4	0	-	✓	-	-
	DLOCATION	Varchar2	30	-	-	-	✓	-	-

1-2

2d

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PROJECT01	PNUMBER	Number	-	4	0	1	-	-	-
	PNAME	Varchar2	30	-	-	-	-	-	-
	PLOCATION	Varchar2	50	-	-	-	✓	-	-
	DNO	Number	-	4	0	-	✓	-	-

1-4

2e

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
WORKSON01	ESSN	Number	-	4	0	-	✓	-	-
	PNO	Number	-	4	0	-	✓	-	-
	HOURS	Number	-	4	0	-	✓	-	-

1-3

2f

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPENDENT01	ESSN	Number	-	4	0	-	✓	-	-
	DEPENDENTNAME	Varchar2	40	-	-	-	✓	-	-
	SEX	Varchar2	20	-	-	-	✓	-	-
	BDATE	Date	7	-	-	-	✓	-	-
	RELATIONSHIP	Varchar2	50	-	-	-	✓	-	-

1-5

```
SEX varchar2(10),
SALARY number(10,2),
SUPERSSN number(4),
DNO number(4) constraint e2 references department01(DNO));
desc employee01;
```

c
```
create table deptLocations01(
DNO number(4) constraint dd1 references department01(DNO),
DLOCATION varchar2(30));
desc deptLocations01;
```

d
```
Create table project01(
PNUMBER number(4) constraint p1 primary key,
PNAME varchar2(30) constraint p2 not null,
PLOCATION varchar2(50),
DNO number(4) constraint p3 references Department01(DNO));
desc project01;
```

e
```
Create table workson01(
ESSN number(4) constraint w1 references employee01(ssN),
PNUMBER number(4) constraint w2 references project01(PNUMBER),
HOURS number(4));
desc workson01;
```

f
```
Create table dependent01(
ESSN number(4) constraint ddd1 references employee01(SSN),
DEPENDENTNAME varchar2(40),
SEX varchar2(20),
BDATE date; RELATIONSHIP varchar2(50));
desc dependent01;
```

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2g)

DNO	DNAME	MGRSSN	MGRSTARTDATE
1	Accounts	101	12-JAN-04
2	Research	102	15-MAR-05
3	Administration	103	20-JUN-06
4	IT	104	10-AUG-07
5	Finance	105	05-SEP-08

2h)

SSN	FNAME	MINT	LNAME	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
101	John	A	Smith	15-FEB-85	Bangalore	Male	55000	—	1
102	Robert	B	Brown	20-MAR-88	Chennai	Male	62000	101	2
103	Mary	C	Jones	10-JUL-90	Mumbai	Female	45000	101	1
104	David	D	Wilson	25-AUG-92	Delhi	Male	48000	102	2
105	Jenifer	E	Taylor	12-SEP-95	Hyderabad	Female	52000	102	2
106	Michael	F	Anderson	05-JAN-89	Pune	Male	38000	103	3
107	Linda	G	Thomas	18-NOV-91	Bangalore	Female	42000	104	4
108	James	H	Jackson	22-APR-94	Mysore	Male	46000	105	5
109	Patricia	I	White	30-JUN-96	chennai	Female	41000	105	5
110	Daniel	J	Harris	14-OCT-93	Bangalore	Male	58000	105	5

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g
```
INSERT INTO DEPARTMENT01 VALUES(1,'Accounts',101,'12-JAN-2004');
INSERT INTO DEPARTMENT01 VALUES(2,'Research',102,'15-MAR-2005');
INSERT INTO DEPARTMENT01 VALUES(3,'Administration',103,'20-JUN-2006');
INSERT INTO DEPARTMENT01 VALUES(4,'IT',104,'10-AUG-2007');
INSERT INTO DEPARTMENT01 VALUES(5,'Finance',105,'05-SEP-2008');
```
→ `Select * from department01;`

h
```
INSERT INTO EMPLOYEE01 VALUES(101,'John','A','Smith','15-FEB-1985',
'Bangalore','Male',55000,NULL,1);
INSERT INTO EMPLOYEE01 VALUES(102,'Robert','B','Brown','20-MAR-1988',
'Chennai','Male',62000,101,2);
INSERT INTO EMPLOYEE01 VALUES(103,'Mary','C','Jones','10-JUL-1990',
'Mumbai','Female',45000,101,1);
INSERT INTO EMPLOYEE01 VALUES(104,'David','D','Wilson','25-AUG-1992',
'Delhi','Male',48000,102,2);
INSERT INTO EMPLOYEE01 VALUES(105,'Jennifer','E','Taylor','12-SEP-1995',
'Hyderabad','Female',52000,102,2);
INSERT INTO EMPLOYEE01 VALUES(106,'Michael','F','Anderson','05-JAN-1987',
'Pune','Male',38000,103,3);
INSERT INTO EMPLOYEE01 VALUES(107,'Linda','G','Thomas','18-NOV-1991',
'Banglore','Female',42000,104,4);
INSERT INTO EMPLOYEE01 VALUES(108,'James','H','Jackson','22-APR-1994'
'Mysore','Male',46000,105,5);
INSERT INTO EMPLOYEE01 VALUES(109,'Patricia','I','White','30-JUN-1996',
'Chennai','Female',41000,105,5);
INSERT INTO EMPLOYEE01 VALUES(110,'Daniel','J','Harris','14-OCT-1993',
'Banglore','Male',58000,105,5);
```
→ `Select * from employee01;`

i
```
INSERT INTO DEPTLOCATIONS01 VALUES(1,'Bangalore');
```

Teacher's Signature

2 i)

DNO	DLOCATION
1	Bangalore
2	Chennai
3	Mumbai
4	Hyderabad
5	Delhi

2 j)

PNUMBER	PNAME	PLOCATION	DNO
1101	Payroll System	Bangalore	1
1102	Research Database	Chennai	2
1103	Employee Management	Mumbai	3
1104	Network Upgrade	Hyderabad	4
1105	Financial Analysis	Delhi	5

2 k)

ESSN	PNO	HOURS
101	1101	20
103	1101	15
102	1102	25
104	1102	20
105	1102	15
106	1103	30
101	1103	10
107	1104	25
102	1104	20
105	1104	15
108	1105	20
109	1105	25
110	1105	30
104	1105	10
101	1105	10

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```
INSERT INTO DEPTLOCATIONS01 VALUES (2, 'Chennai');
INSERT INTO DEPTLOCATIONS01 VALUES (3, 'Mumbai');
INSERT INTO DEPTLOCATIONS01 VALUES (4, 'Hyderabad');
INSERT INTO DEPTLOCATIONS01 VALUES (5, 'Delhi');
```

=> Select * from deptLocations01;

```
j INSERT INTO PROJECT01 VALUES (1101, 'Payroll System', 'Bangalore', 1);
INSERT INTO PROJECT01 VALUES (1102, 'Research Database', 'Chennai', 2);
INSERT INTO PROJECT01 VALUES (1103, 'Employee Management', 'Mumbai', 3);
INSERT INTO PROJECT01 VALUES (1104, 'Network Upgrade', 'Hyderabad', 4);
INSERT INTO PROJECT01 VALUES (1105, 'Financial Analysis', 'Delhi', 5);
```

=> Select * from project01;

```
K INSERT INTO WORKSON01 VALUES (101, 1101, 20);
INSERT INTO WORKSON01 VALUES (103, 1101, 15);
INSERT INTO WORKSON01 VALUES (102, 1102, 25);
INSERT INTO WORKSON01 VALUES (104, 1102, 20);
INSERT INTO WORKSON01 VALUES (105, 1103, 15);
INSERT INTO WORKSON01 VALUES (106, 1103, 30);
INSERT INTO WORKSON01 VALUES (01, 1103, 10);
INSERT INTO WORKSON01 VALUES (107, 1104, 25);
INSERT INTO WORKSON01 VALUES (102, 1104, 20);
INCERT INTO WORKSON01 VALUES (105, 1104, 15);
INSERT INTO WORKSON01 VALUES (108, 1105, 20);
INSERT INTO WORKSON01 VALUES (109, 1105, 25);
INSERT INTO WORKSON01 VALUES (110, 1105, 30);
INSERT INTO WORKSON01 VALUES (104, 1105, 10);
INSERT INTO WORKSON01 VALUES (101, 105, 10);
```

-> Select * from worksON01;

Teacher's Signature

2)

ESSN	DEPENDENTNAME	SEX	BDATE	RELATIONSHIP
101	Alice Smith	Female	10-MAY-10	Daughter
102	Brian Brown	Male	15-JUN-12	Son
103	Carol Jones	Female	20-AUG-11	Daughter
105	Kevin Taylor	Male	05-SEP-15	Son
108	Emma Jackson	Female	12-DEC-18	Daughter

4)i)

SUM(SALARY)	MAX(SALARY)	MIN(SALARY)	AVG(SALARY)
487000	62000	38000	48700

4)ii)

COUNT(*)
10

4)iii)

COUNT(*)
3

4)iv)

DNO	AVG(SALARY)	COUNT(*)
1	50000	2
2	54000	3
3	38000	1
4	42000	1
5	48333.33333333	3

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k
```
INSERT INTO DEPENDENT01 VALUES (101, 'Alice Smith', 'Female',
'10-MAY-2010', 'Daughter');
INSERT INTO DEPENDENT01 VALUES(102, 'Brian Brown', 'Male',
'15-JUN-2012', 'Son');
INSERT INTO DEPENDENT01 VALUES(103, 'Carol Jones', 'Female',
'20-AUG-2011', 'Daughter');
INSERT INTO DEPENDENT01 VALUES (105, 'Kevin Taylor', 'Male',
'05-SEP-2015', 'Son');
INSERT INTO DEPENDENT01 VALUES (108, 'Emma Jackson', 'Female',
'12-DEC-2018', 'Daughter');
```

→
```
Select * from dependent01;
```

4 For a given set of relation schemes, create tables and perform the following simple Queries, Simple Queries with Aggregate functions, Queries with Aggregate functions (group by and having clause)

i Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary.

→
```
Select sum(salary), max(salary), min(salary), avg(salary) from
employee01;
```

ii Retrieve the total number of employees in the company

→
```
Select count(*) from employee01;
```

iii Retrieve the number of employees in the 'Research' department

→
```
Select count(*) from employee01 where DNO=
(select DNO from department01 where DNAME='Research')
group by DNO;
```

iv For each department, retrieve the department number, the number of employees in the department, and their average salary.

Teacher's Signature

4 v)

DNO	AVG(SALARY)
2	54000.

5a)

SSN	SALARY	(SALARY*10)/100 + SALARY
102	62000	68200
104	48000	52800
105	52000	57200

5b)

SUM(SALARY)	AVG(SALARY)	MAX(SALARY)	MIN(SALARY)
100000	50000	55000	45000

6a)

FNAME	MINT	LNAME
James	H	Jaekson
Daniel	J	Harris
John	A	Smith
Patricia	I	White
David	D	Wilson.

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→ Select DNO, avg(salary), count(*) from employee01 group by dno;

v Get all the department number and average salary where average salary is greater than 50000.

→ Select DNO, avg (salary) from employee01 groupby dno having avg (salary) > 50000;

5 Execute the following queries

a. How the resulting salaries if every employee working on the 'Research' Departments is given a 10% raise

→ Select SSN, salary, (salary *10)/100+ salary from employee01, department01 where employee01.dno = department01.dno and dname = 'Research';

6 Find the sum of the salaries of all employees of the 'Accounts' department, as well as the maximum salary, the minimum salary and the average salary in this department.

→ Select sum(salary), avg (salary), max(salary), min(salary) from employee01, department01 where employee01.dno=department01.dno and dname = 'Accounts';

6 Execute the following queries

a. Retrieve the name of each employee controlled by department number 5(use Exists operator).

→ Select fname, minit, lname from employee01 where exists (select pnumber from project01 where dno=5 and pnumber in (select pno from works on01 where essn = ssn));

6) Retrieve the name of each dept and number of employees working

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6b)

DNAME	COUNT(*)
Accounts	2
Research	3
Finance	3

7a)

PNUMBER	PNAME	COUNT(*)
1101	Payroll System	2
1102	Research Database	3
1103	Employee Management	2
1104	Network Upgrade	3
1105	Financial Analysis	5

7b)

FNAME	MINT	LNAME
Mary	C	Jones.

8)

DNAME	COUNT(*)
Accounts	2
Research	3
Finance	3

9)

PNUMBER	PNAME	COUNT(*)
102	Research Database	3
104	Network Upgrade	3
105	Financial Analysis	5

in the each department which has at least 2 employees.

→ Select dname, count(*) from employee01, department01 where employee01.dno = department01.dno group by dname having count(*)>=2;

7. Execute the following queries

a. For each project, retrieve the project number, the project name, and the number of employee who work on that project (use GROUP BY)

→ Select pnumber, pname, count(*) from project01, workson01 where project01.pnumber = workson01.pno group by pnumber, pname;

b. Retrieve the name of employees who born in the year 1990's

→ Select fname, minit, lname from employee01 where to_char(bdate,'YYYY') like '%1990';

8. For each department that has more than five employees, retrieve the department number and number of employees who are making salary more than 40000.

→ Select dname, count(*) from employee01, department01 where employee01.dno = department01.dno and salary > 40000 group by dname having count(*)>=2;

9. For each project on which more than two employees work, retrieve the project number, project name and the number of employees who work on that project.

→ Select pnumber, pname, count(*) from project01, workson01 where project01.pnumber = workson01.pno group by pnumber, pname having count(*)>2;

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10) i)

FNAME	MINT	LNAME	PNAME	HOURS
John	A	Smith	Payroll System	20
Mary	C	Jones	Payroll System	15
Robert	B	Brown	Research Database	25
David	D	Wilson	Research Database	20
Jennifer	E	Taylor	Research Database	15
Michael	F	Anderson	Employee Management	30
Linda	G	Thomas	Network Upgrade	25
Robert	B	Brown	Network Upgrade	20
Jennifer	E	Taylor	Network Upgrade	15
James	H	Jackson	Financial Analysis	20
Patricia	I	White	Financial Analysis	25
Daniel	J	Harris	Financial Analysis	30

10) ii) view dropped.

10) iii)

SSN	FNAME	MINT	LNAME	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
108	James	H	Jackson	22-APR-94	Mysore	Male	46000	105	5
109	Patricia	I	White	30-JUN-96	Chennai	Female	41000	105	5
110	Daniel	J	Harris	14-OCT-93	Bangalore	Male	58000	105	5

3 a i)

Table	Column	Data type	Length	Precision	Scale	Primary Key	Nullable	Default content
DEPARTMENT01	DNO	Number	-	4	0	1	-	-
	DNAME	Varchar2	50	-	-	-	-	-
	MGRSSN	Number	-	4	0	-	✓	-
	MGRSTARTDATE	Date	7	-	-	-	✓	-

3 ii) Table truncated.

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10 For a given set of relation Tables perform the following

a. Creating Views (with and without check option), Dropping views, Selecting from a view.

i -> create view v1 as
(select fname, minit, lname, pname, hours. from employee01, works on 01, project 01 where
employee01.ssn = works on 01.essn and project01.pnumber = works on 01.pno);

-> Select * from v1 where hours > 10;

ii -> drop view v1;

iii -> create view v2 as select * from employee 01 where dno = 5 with check option;

-> select * from v2;

3 Perform the following:

a) Altering a Table, Dropping / Truncating / Renaming Tables, Backing up / Restoring a database.

i Alter table department01 modify dname varchar2(50);

-> desc department01;

ii truncate table dependent01;

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3 iii)

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPENDENTS01	BSSN	Number	-	4	0	-	✓	-	-
	DEPENDENTNAME	Varchar2	40	-	-	-	✓	-	-
	SEX	Varchar2	20	-	-	-	✓	-	-
	BDATE	Date	7	-	-	-	✓	-	-
	RELATIONSHIP	Varchar2	50	-	-	-	✓	-	-

1-5

3 iv) Table dropped.

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iii) Alter table dependent01 rename to dependents01;
desc dependents01;

iv) drop table dependents01;

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